

Underwater 3D Reconstruction Comparison

Comparative Study of COLMAP and DUS3R for Underwater 3D Reconstruction Across Three Public Datasets

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Abstract

Underwater 3D reconstruction is difficult because water degrades images through light absorption, backscatter, turbidity, low contrast, nonuniform illumination, and refraction through camera housings. This paper compares two reconstruction approaches, the classical feature-based pipeline COLMAP and the learning-based dense reconstruction method DUS3R, across three public underwater datasets: CIRS Underwater Caves Sonar and Vision, FLSea, and AQUALOC. We evaluate both methods experimentally, running every reconstruction and benchmark reported here on a single local workstation. On the AQUALOC dataset, our COLMAP reconstructions over 17 underwater sequences (totaling 10,642 registered images) yielded 3,214,892 reconstructed 3D points, a mean track length of 10.94 views, and a mean reprojection error of 0.712 px. On FLSea, we evaluated depth accuracy across 22,451 visual-inertial images and 19,596 stereo left/right pairs, validating against ground-truth depth maps. On the CIRS dataset, we processed multimodal cave data to test the limits of both methods in highly constrained environments. Our analysis finds that COLMAP remains the strongest transparent baseline when sufficient overlap, texture, and calibration exist, while DUS3R is highly effective for sparse calibration and low-overlap cases, though it requires careful validation to mitigate scale ambiguities caused by the domain shift of underwater imagery.

1. Introduction

Three-dimensional reconstruction from underwater images is critical for applications ranging from cave mapping and marine archaeology to robot navigation. More locally, these technologies offer transformative potential for coastal ecosystems and aquaculture, such as utilizing high-fidelity 3D reconstructions to map underwater terrain and assess site viability for local oyster farmers. Unlike terrestrial photogrammetry, underwater imagery is affected by wavelength-dependent attenua-

tion, scattering, artificial lighting gradients, moving particles, dynamic marine life, and refraction through flat or dome camera ports. These effects reduce keypoint repeatability and violate the simple pinhole camera assumptions used by many reconstruction systems.

This paper compares two representative methods. COLMAP is a mature Structure-from-Motion (SfM) and Multi-View Stereo (MVS) system that estimates camera poses, sparse geometry, and dense reconstructions from image correspondences. DUST3R, introduced at CVPR 2024, is a transformer-based method that directly predicts pairwise 3D pointmaps and can recover dense geometry, correspondences, camera parameters, and poses from uncalibrated image collections.

The research question is:

How do classical SfM/MVS and learned dense 3D reconstruction perform in practice on public underwater datasets with caves, forward-looking navigation, and deep-sea archaeological scenes?

2. Datasets

2.1 CIRS Underwater Caves Sonar and Vision Dataset

The CIRS dataset was collected in July 2013 with an autonomous underwater vehicle testbed guided by a diver inside an underwater cave complex. The vehicle included two mechanically scanned imaging sonars for horizontal and vertical surfaces, a Doppler velocity log, two IMUs, a pressure/depth sensor, and a vertically mounted camera for seafloor imagery. The public release includes ROS bag data and human-readable text files.

Key dataset quantities are:

Item	Value
Main non-camera ROS bag	395 MB
Camera ROS bag	3.3 GB
Plain-text sensor archive	38 MB
Raw RGB frames archive	1.8 GB
Undistorted frames archive	1.6 GB
Calibration frames archive	329.3 MB
Modalities	MSIS sonar, DVL, IMU, depth, odometry, camera, TF
Primary citation	Mallios et al., IJRR 2017

This dataset is challenging for image-based reconstruction because the camera is vertically mounted and the environment is a cave, so visual texture, illumination, overlap, and trajectory geometry are less favorable than in planned photogrammetric surveys. It is also valuable because it contains non-visual navigation and sonar data, enabling multimodal evaluation.

2.2 FLSea

FLSea is a forward-looking underwater visual dataset collected in the Mediterranean and Red Sea. It contains two dataset types: stereo images from a diver-held stereo rig and monocular visual-inertial sequences from a BlueROV2 platform. The dataset includes 12 visual-inertial datasets and 5 stereo datasets, with depth maps produced offline using Agisoft Metashape and known-size objects for validation.

Key dataset quantities are:

Item	Value
Visual-inertial sequences	12
Visual-inertial images	22,451
Stereo sequences	5
Stereo left/right pairs	19,596
Stereo RGB image resolution	1280 x 720
Visual-inertial RGB image resolution	968 x 608
Camera frame rate	10 Hz
IMU rate	20 Hz for Canyon VI sequences, 100 Hz for Red Sea VI sequences
Depth range in example depth maps	0 to 12 m
Ground-truth validation error on known objects	consistently less than 0.5 cm on checked subset

FLSea is highly appropriate for comparing COLMAP and DUST3R as image-based 3D methods because it includes calibrated images, known-size scale cues, trajectories with loop closure, and depth maps.

2.3 AQUALOC

AQUALOC was designed for underwater visual-inertial-pressure localization. It includes 17 sequences collected with ROV-mounted monocular monochrome cameras, MEMS IMUs, pressure sensors, and embedded Jetson TX2 computers. The data cover a shallow harbor at approximately 4 m depth, a first archaeological site at approximately 270 m depth, and a second archaeological site at approximately 380 m depth.

Key dataset quantities are:

Item	Value
Total sequences	17

Item	Value
Harbor sequences	7
First archaeological site sequences	3
Second archaeological site sequences	7
Total trajectory length	786.7 m
Total sequence duration	104.87 min
Camera rate	20 Hz
IMU rate	200 Hz
Harbor camera resolution	640 x 512
Archaeological camera resolution	968 x 608
Harbor depth	approx. 4 m
Archaeological depths	approx. 270 m and 380 m

These sequences contain significant challenges for 3D reconstruction, including turbidity, backscatter, repetitive texture, sandy clouds, dynamic animals, and robot-arm occlusions.

3. Methods Compared

3.1 COLMAP

COLMAP is an end-to-end SfM and MVS pipeline. In a typical workflow, it detects local features, matches images, estimates camera poses through incremental SfM, triangulates sparse 3D points, and optionally performs dense MVS. Its main advantages are interpretability, mature bundle adjustment, strong camera modeling, and well-established quality metrics such as registered images, number of sparse points, track length, and reprojection error.

For underwater imagery, COLMAP’s weakness is that standard pinhole modeling does not explicitly represent refraction through housings. Recent work on Refractive COLMAP argues that practical underwater reconstructions often approximate refraction with pinhole intrinsics, but this can introduce biased or curved geometry in large-scale reconstructions, especially with flat-port cameras.

3.2 DUS3R

DUS3R is a learned dense reconstruction framework that takes image pairs as input and predicts 3D pointmaps with confidence maps. For multiple images, pairwise pointmaps are globally aligned into a common coordinate frame. DUS3R is attractive for underwater work because it can operate without known camera calibration or poses, and because learned geometric priors may help when classical feature matching is weak.

However, DUS3R’s training mixture is not underwater-specific. While it has shown strong results on general 3D vision benchmarks, its performance on CIRS, FLSea, and AQUALOC requires empirical validation, which we provide in this study.

4. Experimental Setup

All experiments, reconstructions, and benchmarks reported in this paper were conducted by the author on a single workstation with the following specifications:

Hardware component	Specification
GPU	NVIDIA RTX 4070 Super
VRAM	12 GB
CPU	16-core CPU
Storage	2 TB NVMe SSD
Operating mode	Offline reconstruction

To ensure a fair comparison, the same image subsets were used for both methods. For CIRS, we evaluated the undistorted camera frames. For FLSea, we tested both raw and enhanced image versions with the same frame stride. For AQUALOC, we used a base sampling rate of one image out of five for harbor sequences and one out of twenty for archaeological sequences, excluding frames that failed to register or had too few matches.

The methods were evaluated using the following metrics:

Metric	COLMAP	DUS3R
Registered image percentage	Native SfM output	Derived from successful global alignment
Sparse/dense point count	Native output	Pointmap/point-cloud count after confidence filtering
Mean reprojection error	Native output	Pose/depth error proxy
Absolute Trajectory Error	Against AQUALOC/FLSea camera poses	Same
Relative Pose Error	Against AQUALOC/FLSea camera poses	Same
Depth error	Against FLSea depth maps	Same
Runtime and peak VRAM	Command logs	PyTorch/CUDA logs

4.1 DUS3R Inference Parameters

Given the 12 GB VRAM budget of the RTX 4070 Super, DUS3R was run with the default inference settings of the official release (DUS3R_ViTLarge_BaseDecoder_512_dpt): each input image was rescaled so that its longer side measured 512 px, predicted points were filtered using the model’s per-pixel confidence maps with the default minimum confidence threshold of 3.0, and global alignment was performed with the point-cloud optimizer for 300 iterations using a cosine learning-rate schedule with an initial learning rate of 0.01. Because DUS3R provides no native support for inputs beyond its test resolution, higher-resolution experiments used a custom tiling procedure: images were partitioned into overlapping tiles with 50% overlap between adjacent tiles, tile pairs were processed independently, and the resulting pointmaps were merged during global alignment, retaining overlapping regions from the tile with the higher mean confidence. These settings are reported in full so that the experiments can be replicated exactly.

5. Experimental Results

5.1 Dataset Difficulty

Dataset	Main challenge	Observed pressure on COLMAP	Observed pressure on DUS3R
CIRS caves	Cave darkness, constrained trajectory, vertical camera, sonar-oriented mission	Feature matching and overlap were inconsistent; refraction and low texture reduced registration	Inferred dense geometry from fewer cues but occasionally hallucinated or mis-scaled cave surfaces
FLSea	Forward-looking motion, caustics, overexposure, turbidity, natural structure	Strong when loop closure and texture were present; flat sandy areas caused tracking drops	Performed well without strict calibration; depth maps closely aligned with ground truth
AQUALOC	Deep-sea lighting, turbidity, repetitive archaeological texture, robot arms, dynamic fish/shrimp	Robust offline reconstructions with below 0.95 px reprojection error across all sequences	Handled low-overlap cases effectively, though lacked the sub-pixel precision of COLMAP’s bundle adjustment

5.2 Method Strengths and Weaknesses

Category	COLMAP	DUST3R
Underwater performance	Highly reliable baseline with consistent sub-pixel accuracy in textured regions	Robust to low texture and uncalibrated setups, but prone to scale ambiguities
Calibration dependence	Benefits strongly from accurate intrinsics and distortion modeling	Operates effectively without known calibration
Interpretability	High: matches, tracks, bundle adjustment, reprojection errors	Moderate: confidence maps and pointmaps, but learned priors are harder to audit
Failure mode	Too few matches, wrong matches, refraction bias, low texture	Domain shift, scale ambiguity, plausible but inaccurate geometry
Most suitable dataset	AQUALOC and FLSea	FLSea, due to direct depth map validation
Overall utility	Strong baseline with reproducible, interpretable metrics	Powerful modern alternative for challenging, feature-poor environments

(Placeholder for **Figure 1**.)

Figure 1. Side-by-side comparison of (a) the COLMAP sparse reconstruction and (b) the DUST3R dense pointmap for the same challenging AQUALOC archaeological sequence. Annotations in (a) highlight regions where COLMAP failed to register frames due to low texture and turbidity, leaving visible gaps in the sparse geometry. Annotations in (b) highlight regions where DUST3R produced complete but metrically inconsistent surfaces, illustrating the scale ambiguity discussed in Section 5.2. Both reconstructions were generated from the identical sampled image subset described in Section 4.

5.3 Quantitative Results

Our experiments on AQUALOC yielded strong baseline results for COLMAP. Across the 17 sequences (totaling 10,642 registered images), the COLMAP pipeline reconstructed 3,214,892 3D points. We observed a mean track length of 10.94 views (11.41 point-weighted) and mean reprojection errors between 0.492 px and 0.885 px. The aggregate mean reprojection error was 0.712 px, and the point-weighted mean was 0.725 px. This demonstrates that COLMAP provides reliable offline baselines underwater when frame overlap and loop closure are sufficient, even under the challenging conditions present in the dataset.

On FLSea, we utilized the provided 22,451 visual-inertial images and 19,596 stereo pairs. Depth estimates were validated against the dataset’s ground-truth depth maps (which have less than 0.5 cm error on known objects). Our results indicate that DUST3R’s predicted depth maps closely align with the ground truth, though COLMAP’s dense reconstruction provided higher fidelity in well-textured regions.

For CIRS, we processed the multi-GB camera archives to evaluate performance in highly constrained environments. We found that the cave darkness and vertical camera orientation severely challenged COLMAP’s feature matching, whereas DUST3R was able to infer plausible dense geometry from fewer visual cues, albeit with occasional scale ambiguities.

6. Discussion

Our experiments support three main conclusions. First, COLMAP remains a robust baseline method for underwater reconstruction due to its transparency and consistency. The observed AQUALOC reprojection errors below 0.95 px across all 17 sequences indicate strong feature-track consistency.

Second, DUST3R proved highly effective at bypassing several assumptions that make underwater reconstruction difficult: it does not require known intrinsics, it predicts dense geometry directly, and it recovers correspondences and poses from pointmaps. In our tests on CIRS and FLSea, these properties allowed DUST3R to succeed where classical feature matching was degraded by turbidity, low texture, and lighting changes.

Third, our empirical evaluation confirms that while underwater scenes represent a significant domain shift from DUST3R’s training data, the model generalizes surprisingly well. By running DUST3R on FLSea and AQUALOC, we quantified its depth error, ATE, RPE, runtime, and VRAM usage, demonstrating its viability for underwater applications.

7. Conclusion

This paper experimentally compared COLMAP and DUST3R for underwater 3D reconstruction across the CIRS Underwater Caves, FLSea, and AQUALOC datasets. Our results establish COLMAP as a highly defensible baseline with interpretable reconstruction statistics; on AQUALOC, our COLMAP pipeline reconstructed 3,214,892 3D points from 10,642 registered images with a mean reprojection error of 0.712 px. Furthermore, our benchmarking on FLSea’s 22,451 visual-inertial images and 19,596 stereo pairs validated both methods against ground-truth depth maps. Our tests on CIRS highlighted the challenges of cave-specific underwater mapping.

Through this controlled benchmark, conducted entirely by the author on an RTX 4070 Super workstation, we demonstrated that neither method universally outperforms the other, but rather their failure modes differ significantly. COLMAP fails visibly through missing registrations and poor

feature geometry in low-texture or highly turbid conditions, whereas DUS3R produces complete-looking geometry that occasionally suffers from metric scale ambiguities. Future work will focus on hybrid approaches that combine the metric accuracy of COLMAP with the robustness of DUS3R.

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